Neural Structural Reasoner: A Brain-inspired Architecture for Reasoning over Structured Knowledge

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Abstract

Structural reasoning, the ability to recognize and make inferences over the relational structure between objects and concepts, is a hallmark of human cognition, yet prevailing methods often collapse relational topology into flat embeddings, cannot discover hidden structure and lack interpretability. We introduce Neural Structural Reasoner (NSR), a brain-inspired network that preserves relational structure directly in the connectivity and dynamics of coupled neuronal populations. NSR draws inspiration from three biological mechanisms: multi-layered architecture for encoding hierarchical knowledge, stable representations of entity and concepts, and path integration for input-driven state inference. At query time, NSR parallelizes computation over candidate relational structures and leverages confidence-weighted scores to perform link prediction. Across standard knowledge-graph benchmarks, NSR achieves competitive accuracy without leading on every dataset, and has lower reported training times than several neural baselines. Because reasoning is implemented through sequences of human-readable neuron activations, NSR affords native interpretability by tracking intermediate inference steps. The model further extracts latent relational hierarchies and compositional rules, demonstrating the brain-inspired architecture as an effective, efficient, and highly interpretable substrate for structural reasoning.

1 Introduction

Knowledge in the real world is inherently structured—organized by relations such as order, hierarchy, and connectivity—and humans routinely rely on these relations, rather than the entities alone, to reason about the world [19, 22]. This ability to understand and reason about data relationships, which we call *structural reasoning*, has long been regarded as a desideratum for human-like machine intelligence [27, 55]. Machine learning pursues it from several angles: knowledge-graph embeddings map entities and typed relations into continuous spaces for multi-hop queries [24, 25, 54];graph neural networks impose relational inductive biases via message passing [4, 43, 52]; and large language models, now widely deployed as general reasoners, tackle structural tasks by verbalizing relations and chaining inferences in natural language [59, 23, 17, 37]. Despite their differences, these methods share two recurring weaknesses: they tend to collapse rich structural information into flat

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vector representations and produce answers without exposing how they were reached—making their reasoning hard to verify and prone to break on deeper, multi-step queries [3, 61].

Biology offers an alternative starting point. Structural reasoning in the brain arises from the coordinated activity of several systems: the hippocampal–entorhinal circuit builds cognitive maps—relational organizations of entities that support inference beyond direct experience across spatial and non-spatial domains [5, 57], and does so predictively and in a manner causally required for structure-based inference [46, 34]; the prefrontal cortex abstracts relations into hierarchical, schema-like representations that generalize across contexts [42].Existing biologically inspired models, including the Tolman–Eichenbaum Machine [57], Vector-HaSH [10], and related associative-memory architectures [47, 49, 41], draw on these neural principles to represent and retrieve structured knowledge. However, to our knowledge, there still remains a lack of a modular, brain-inspired architecture that can both encode and perform reasoning over structured knowledge at scale.

We address this gap with Neural Structural Reasoner (NSR), a brain-inspired network architecture for structural reasoning. The main advances are four-fold:

- **Network structure as relational structure.** NSR uses network connectivity to represent relational structure, and exploits efficient network dynamics to implement relational reasoning.
- **Parallel reasoning with path-support scores.** NSR computes in parallel over multiple candidate relational structures, returning ranked candidate answers with confidence score.
- **Competitive performance with high computational efficiency.** NSR achieves competitive, dataset-dependent accuracy across benchmarks while requiring relatively less training time.
- **Interpretability from dynamics.** NSR yields human-readable reasoning paths by tracking network dynamics as reasoning steps at inference.

2 Background and Related Works

2.1 Structural reasoning across machine learning

Structural reasoning has long been studied under Knowledge Representation and Reasoning (KRR) [7], which formalizes knowledge through structured objects—logic rules, ontologies, and graphs—and defines inference as a process operating on that structure. Among these, knowledge graphs (KGs) offer an instantiation that is both scalable in practice and conducive to systematic empirical study. Formally, a KG is a tuple $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ with entities $\mathcal{E}$, relations $\mathcal{R}$, and triples $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$, where each $(h, r, t) \in \mathcal{T}$ asserts that entity h stands in relation r to entity t [24, 25]. The canonical task is link prediction: given a query $(h, r, ?)$, score candidate tails and recover the missing entity from the relational structure already in $\mathcal{G}$. This makes structural reasoning concrete—inference over typed relations among discrete entities, isolated from perceptual or linguistic confounds.

A central object of study is the structure of relations themselves, since real KGs are pervaded by canonical patterns like symmetry (e.g., `sibling_of`), inversion ($r_1(x, y) \Rightarrow r_2(y, x)$, e.g., `teacher_of` $\leftrightarrow$ `student_of`), and composition ($r_1(x, y) \wedge r_2(y, z) \Rightarrow r_3(x, z)$, e.g., `father_of` $\circ$ `father_of` $\Rightarrow$ `grandfather_of`) [48]. Composition is especially important: nearly every relation in benchmark KGs participates in some compositional pattern [32]. Existing methods address these regularities through distinct representational strategies, each with characteristic limitations. **Geometric embedding and tensor-factorization models** encode relations as translations, rotations, bilinear maps, or low-rank interactions, enabling scalable link prediction but leaving composition implicit in vector geometry [6, 48, 30, 51]. **Deep neural models**, including ConvE [15] and message-passing GNNs [44, 62, 21], add nonlinear computation and neighborhood aggregation, but often lack transparent inference trajectories and struggle with long-range relational chains. **Rule-mining and path-based systems** expose logical chains, but rely on combinatorial search and are brittle under missing or noisy edges [60, 14]. LLM-based methods can verbalize triples or retrieve subgraphs, yet their outputs are prompt-sensitive and may not faithfully track executable graph operations. Thus, current methods generally lack a native mechanism for representing relational structure as explicit, reusable, and traceable dynamics over a structured network.

2.2 Neural substrates of structural reasoning

Structural reasoning in the brain appears to rely on a distributed network of brain regions. Conceptual knowledge is often linked to the anterior temporal lobe (ATL), a proposed transmodal semantic hub that integrates modality-specific features into **stable, context-general concepts** [36, 40]. Relational operations recruit prefrontal mechanisms, especially rostrolateral/frontopolar prefrontal cortex, whose **activity increases with relational integration** demands and which is implicated in analogical and multi-relational reasoning [12, 8, 56, 53]. **Flexible composition over relations** further depends on medial temporal lobe circuitry: the hippocampus binds overlapping experiences into relational structures that support novel inference [16, 38, 18], while entorhinal grid-like codes may provide a metric format for cognitive maps spanning physical and conceptual spaces [13, 35]. Through **path integration**, hippocampal–entorhinal circuits update an internal state estimate by accumulating transitions through space [5]; by extension, analogous dynamics may support abstract reasoning by tracking trajectories through relational or conceptual spaces (though direct evidence remains scarce). Thus, structural reasoning in the brain likely emerges from specialized but interconnected regions that support stable entity and relation representations, relation-specific binding of entities, and compositional generalization of relational structures.

2.3 Biologically inspired models of structural reasoning

Some biologically inspired architectures offer important ingredients for relational cognition, though they have not yet fully addressed structural reasoning in the sense of §2.1. The Tolman–Eichenbaum Machine (TEM) [57, 58] repurposes hippocampal–entorhinal path integration for abstract inference, treating relations as velocity-like signals over a learned structural scaffold; however, its entorhinal module is implemented as a backpropagation-trained RNN, leaving open how such computations arise from grid-cell-like dynamics. Vector–HaSH [10] provides a more biologically grounded HPC–EC loop and supports sequence memory via path integration, but its role in abstract relational reasoning remains unexplored. Continuous-attractor models of grid cells [9] offer a mechanistic account in which velocity-related inputs drive state transition to encode movement in space, yet their application is often limited to low-dimensional navigation rather than high-dimensional relational domains. A deeper limitation of many HPC-EC inspired models is that the network dynamics, built for 2D navigation, are confined to a 2D torus, while **abstract relations require input-driven transitions on high-dimensional manifolds**—and no biologically plausible architecture have been shown to supports this. Together, these works motivate a bio-inspired architecture that can encode and perform reasoning in structured, high-dimensional relation space.

3 The Neural Structural Reasoner Model

3.1 Model Architecture

Neural Structural Reasoner (NSR) maps relational structure between entities onto a hierarchical multi-layered network, where connectivity directly encodes associations between entities, relations and relation compositions (Fig. 1a). The core functional layers are:

Entity Layer (L_E): contains neurons $\{e_i\}^N$ encoding the N entities in the data.

Relations Layer (L_R): contains a total of $2M$ neurons where the first half, $\{r_i\}^M$, encode the M relations and the second half, $\{r_{i+M}\}^M$, encode the inverse relations $\{r^{-1}\}^M$ (e.g., $\texttt{father_of}^{-1}$ is learned to associate with `son_of` via the Hebbian rule in §3.3).

Relational Reasoning Layer (L_Z): contains up to $N \times (2M)$ neurons; a given neuron z_i^k encodes the association between the head entity h_i and the relation r_k.

Relation Composition Layer (L_C): contains a flexible number of neurons encoding the compositional structures (e.g. chains) among relations.

We assess the functional contribution of the mechanisms implemented by these layers in Appendix C.5, through ablations of inverse-relation encoding, relation-equivalence retrieval, compositional inference, and Hebbian updates.

3.1.1 Network Connectivity and Dynamics

The four layers are connected in a bi-directional hierarchy: $L_E \leftrightarrow L_Z \leftrightarrow L_R \leftrightarrow L_C$. L_E and L_Z neurons encoding the same entity are bidirectionally coupled at initialization and held fixed:

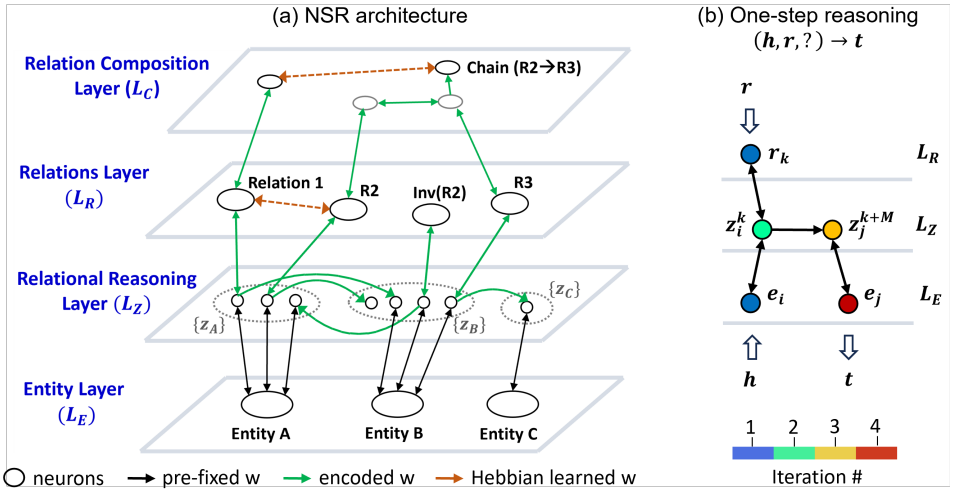


Figure 1: **The Neural Structural Reasoner Architecture.** (a) Four core layers of NSR: L_E encode entities; L_Z instantiates relation-driven state transition; L_R and L_C encode relations and their compositions (e.g., chains). Connections in black are fixed a priori, those in green encode observed triples, those in orange are learned via Hebbian updates to capture latent relational structure. (b) Sequence of neuron activation that implements one-step reasoning over $(h, r, ?) \mapsto t$.

$$\forall i \in [N],\ \forall k \in [2M]: \quad W_{E \leftrightarrow Z}(e_i, z_i^k) = 1 \tag{1}$$

All other weights are initialized at 0 and are learnable. The activation state of each layer is updated in a single step by summing up intra- and inter-layer inputs and external query input I, and transformed through sigmoidal nonlinearity σ.

$$\begin{cases} L_E : X_E(t+1) = \sigma_E(W_{E \leftrightarrow Z} X_Z(t)) + I_E, \\ L_Z : X_Z(t+1) = \sigma_Z\big(W_{ZZ} H\big(X_Z(t) - \eta_Z\big) + W_{Z \leftrightarrow E} X_E(t) + W_{Z \leftrightarrow R} H\big(X_R(t) - \eta_R\big)\big) + I_Z, \\ L_R : X_R(t+1) = \sigma_R\big(W_{RR} X_R(t) + W_{R \leftrightarrow Z} H\big(X_Z(t) - \eta_Z\big) + W_{R \leftrightarrow C} X_C(t)\big) + I_R, \\ L_C : X_C(t+1) = \sigma_C\big(W_{CC} X_C(t) + W_{C \leftrightarrow R} H\big(X_R(t) - \eta_R\big)\big) + I_C \end{cases} \tag{2}$$

Neuronal interaction within L_Z are rectified by the Heaviside function H, ensuring that activity propagates to only in response to convergent inputs from the associated entity and relation. Similar gating exists for projections $L_Z \rightarrow L_R$ and $L_R \rightarrow L_C$ to conditional activation of co-occurring relations and relation compositions in the learning phase (§3.3). When performing reasoning, the relevant entities and relations obtained in intermediate steps can be read out directly from neuronal activations in L_E, L_R or L_C.

3.2 Encoding of Relation Triples

For each triple (h_i, r_k, t_j) and its inverse (t_j, r_k^{-1}, h_i), we let neurons r_k and r_{k+M} to encode the relation and its inverse, and connect them with the entity-encoding neurons by assigning the following weights :

$$\begin{cases} W_{ZZ}(z_i^k, z_j^{k+M}) = W_{ZZ}(z_j^{k+M}, z_i^k) = 1, \\ W_{Z \leftrightarrow R}(z_i^k, r_k) = 1, \quad W_{Z \leftrightarrow R}(z_j^{k+M}, r_{k+M}) = 1 \end{cases} \tag{3}$$

3.2.1 Reasoning over $(h_i, r_k, ?)$ through Path-Integration-Inspired Dynamics

When queried with the relation triple $(h_i, r_k, ?)$ (Fig. 1b), NSR first activates the entity neuron e_i and the relation neuron r_k. Iterating over (2) then activates neurons z_i^k and z_j^{k+M} through intra-layer dynamics in L_Z. This in turn activates neuron e_j, which allows the readout of the tail entity t_j. This computational logic is inspired by path integration models proposed for entorhinal grid cells [9, 10].

3.3 Learning Compositional Rules among Relations

Following the encoding phase, NSR uses Hebbian-like associative learning to extract the following compositional structure and equivalence rules among relations:

Relational Equivalence: When queried with two relation triples with the same head entity, e.g. (h_i, r_k, t_j) and (h_i, r_l, t_w), L_Z neurons encoding the tail entities, z_j^{k+M} and z_w^{l+M} will be activated. If in fact $t_j = t_w$, then the activity of z_j^{k+M} and z_j^{l+M} can be elevated above the threshold η_z through mutual coupling with e_j. Iterating over (2) leads to co-activation of r_{k+M} and r_{l+M} and of r_k and r_l. The connections between co-activated relation neurons are updated via a symmetrized version of Oja's rule:

$$\Delta w_{ij}^{RR} = \delta_{RR}(r_i * r_j) - \mu_{RR}(r_i^2 + r_j^2)w_{ij}^{RR} \tag{4}$$

Where δ_{RR} and μ_{RR} are learning rates. In practice, all relation triples with the same head entity can be queried at once, and weight update can occur simultaneously for all co-activating r_ks. This ultimately leads to strong coupling between neurons encoding semantically analogous relations.

Symmetric and Inverse Relations: Same network dynamics and learning rule can be applied to extract symmetric and inverse relations. This is achieved by letting $r_l = r_l^{-1}$ in the case of symmetry, and $r_l = r_m^{-1}$ in the case of inverse relations.

Composite relations: We focus on detecting the equivalence between multi-hop relation chains and single triples. Take two-hop chains for example, this amounts to detecting

$$r_p(h_i, x_p) \wedge r_q(x_p, x_q) \Rightarrow r_k(h_i, x_q) \Leftrightarrow r_k^{-1}(x_q, h_i). \tag{5}$$

where the bidirectional equivalence follows from Eq. 3: every triple (h, r, t) is encoded together with its inverse (t, r^{-1}, h). The network discovers such rules by checking whether a two-hop chain returns to the original head entity. Concretely, suppose (h_i, r_p, x_p) is already encoded. NSR computes three inference steps:

Step 1: Activate e_{x_p} (the tail of the known triple) and all relation neurons $\{r_m\}_{m=1}^{2M} \setminus \{r_{p+M}\}$ (all relations except the inverse r_p^{-1}). Iterating dynamics in L_Z activates neurons z_q^m representing candidate tails x_q reachable from x_p via each relation r_m.

Step 2: For each discovered entity x_q, treat it as a new head entity and repeat Step 1, querying $(x_q, r_m, ?)$. This yields a second-hop set of tails $\{x_s\}$ encoded by neurons z_s^m.

Step 3: If any x_s coincides with the original head entity h_i, the network has closed a three-step loop:

$$(h_i, r_p, x_p) \wedge (x_p, r_q, x_q) \wedge (x_q, r_k^{-1}, h_i). \tag{6}$$

Because the loop returns to h_i through r_k^{-1}, the two-hop composition $r_p \circ r_q$ connects h_i to x_q in the direction of r_k, supporting the rule $r_p \circ r_q \Rightarrow r_k$ (Eq. 5). Owing to the bidirectional encoding in Eq. 3, the presence of (x_q, r_k^{-1}, h_i) is equivalent to the presence of (h_i, r_k, x_q); hence the discovered chain is registered in L_C as a compositional association between $r_p \to r_q$ and r_k.

Whenever such a closed loop is detected, assign a L_C neuron c_k to represent the relation r_k, and neuron $c_{p\to q}$ to detect the sequential activation of $r_p \to r_q$. Update weights according to:

$$\Delta w_{k-pq}^{CC} = \delta_{CC}(c_k * c_{pq}) - \mu_{CC}(c_k^2 + c_{pq}^2)w_{k-pq}^{CC} \tag{7}$$

While these procedures are designed for online sequential learning, for static datasets, the existence of composite structures can be detected statistically and then encoded directly in network weights. We present details of these simplifications along with pseudocodes in Appendix A.2.

3.4 Reasoning Phase

Given a query $(h_i, r_k, ?)$ where $(h_i, r_k, t^*) \notin \mathcal{D}_{\text{train}}$, NSR uncovers t^* by first retrieving equivalent relations r_e or relation compositions C_e to r_k, and then performing relational reasoning over these equivalence sets to obtain candidate tail entities $\tilde{\mathcal{T}} = \{\tilde{t}\}$ with path-support scores used for ranking:

Step 1 **Retrieval of equivalent relations:** This is achieved by first activating neuron r_k encoding the query relation, and iterating the dynamics of L_R in (2) once to activate neurons $\{r_e\}$ encoding the set of equivalent relations. A threshold `T_thresh` gates the propagation, retaining only neurons with activation at least `T_thresh` ; their activation levels, which represent how strongly two relations are associated, serve as path-support scores (i.e., $S_R(r_e) = x_e$).

Step 2 **Relational reasoning with $\{r_e\}$:** This is implemented by activating the neuron e_i encoding the head entity, while holding $\{r_e\}$ active. Iterating over the dynamics of L_Z according to (2) implements $(h_i, r_e, ?)$ and subsequently activates neurons $\{\tilde{e}\}$ in L_E. These neurons encode a set of candidate tail entities $\tilde{\mathcal{T}}^{(1)} = \{\tilde{t}\}$.

Step 3 **Retrieval of equivalent relation compositions:** Upon re-initializing the network, activate the query relation neuron r_k and allow its activity to propagate to L_C. Iterating L_C dynamics based on (2) activates neurons encoding relation compositions that are closely associated with r_k. Applying a gate function yields the top-K activated neurons $\{c_e\} = \text{TopK}(X_C)$ corresponding to the top equivalent compositions. The activation of $\{c_e\}$ serves as the path-support scores S_C.

Step 4 **Relational reasoning with $\{c_e\}$:** Sample and activate a single neuron in $\{c_e\}$, which then initiates the sequential activation of L_R neurons $r_{p_1}, r_{p_2}, \ldots, r_{p_\ell}$ in the relation chain. Activate the entity neuron e_i together with r_{p_1} to initiate multi-hop reasoning in L_Z. At the end of the chain traversal, retain the contributing grounded paths together with their terminal entities in L_E. Each path receives the support score $S_C(c_e)$ of its corresponding composition neuron. Repeating this procedure for all selected composition neurons yields the compositional path collection and its candidate tail set $\tilde{\mathcal{T}}^{(2)}$. Distinct groundings of the same relation chain are retained separately for aggregation in Step 5.

Step 5 **Aggregating path-support scores for candidate entities:** For each candidate tail entity t_j, let $\mathcal{P}(t_j)$ be the set of grounded equivalent-relation and compositional paths reaching it, and let $s(p)$ be the support score assigned to path p. We compute

$$\text{Score}_a(t_j) = \begin{cases} \max\limits_{p \in \mathcal{P}(t_j)} s(p), & a = \max, \\ \sum\limits_{p \in \mathcal{P}(t_j)} s(p), & a = \text{sum}, \end{cases} \tag{8}$$

where the aggregation mode a is selected on the validation set. Max aggregation may overvalue a single spurious path, whereas sum aggregation may overcount correlated paths. These scores are used to rank candidates, not as calibrated probabilities; the contributing paths remain inspectable.

4 Experiments

4.1 Experimental setup

Datasets. Our main-table evaluation uses four benchmarks: Nations [26], Kinship [27], YAGO3-10 [15], and FB15k-237 [50], accessed through PyKEEN [2]. These benchmarks span compact relational graphs and larger knowledge graphs. We additionally report experiments on two standard benchmarks, Countries S3 [31] and WN18RR [15], and on our constructed Kinship1990_EXTENDED dataset in Appendix C.4. The construction of Kinship1990_EXTENDED is described in Appendix C.2. For the six public benchmarks, all locally evaluated methods use the provided train/validation/test splits.

Evaluation. We adopt the filtered link-prediction setting [6], reporting Mean Reciprocal Rank (MRR) and `Hits@K` ($K \in \{1, 3\}$). In addition to predictive accuracy, we measure training time to assess computational efficiency.

Table 1: Link-prediction performance on Nations and Kinship. Hits@1 and Hits@3 are percentages. Higher ranking scores and lower training times are better; the best value in each column is bold.

		Nations			Kinship			Training time	
Category	Method	MRR	H@1	H@3	MRR	H@1	H@3	Nations	Kinship
Embedding-based	ConvE	0.8029	69.05	89.45	**0.7927**	**68.11**	**88.45**	26 s	64 s
	RotatE	0.5351	33.23	66.02	0.7598	63.37	86.26	24 s	84 s
Symbolic rule learning	AnyBURL	0.7994	69.15	89.55	0.6768	54.10	76.63	64 s	63 s
	AMIE	**0.8559**	**77.11**	**92.54**	0.6767	55.03	76.26	0.6 h	**2 s**
Neural rule learning	NeuralLP	0.6841	52.74	81.59	0.6072	47.30	68.06	40 s	26 s
	NCRL	0.4571	25.37	55.22	0.6050	45.81	68.99	134 s	99 s
	RNNLogic	0.7216	61.65	83.06	0.6690	54.94	77.60	32 s	82 s
Path-based reasoning	PRA / PathRank	0.5933	38.31	77.61	0.6296	46.83	73.56	34 s	86 s
	MINERVA	0.5865	44.28	73.13	0.6253	46.37	73.46	0.9 h	1.2 h
GNN-based	NBFNet	0.7479	62.19	84.08	0.7445	62.38	83.61	29 s	77 s
Brain-inspired	NSR	0.8142	71.64	88.16	0.6515	54.21	70.67	**3 s**	11 s

Table 2: Link-prediction performance on YAGO3-10 and FB15k-237. Hits@1 and Hits@3 are percentages. Higher ranking scores and lower training times are better; the best available value in each column is bold. A dash indicates a missing entry.

		YAGO3-10			FB15k-237			Training time	
Category	Method	MRR	H@1	H@3	MRR	H@1	H@3	YAGO3-10	FB15k-237
Embedding-based	ConvE	**0.6365**	**59.03**	**71.28**	0.4095	31.61	44.89	10.3 h	991 s
	RotatE	0.1812	9.98	21.04	0.3368	27.07	41.51	1.5 h	1.5 h
Symbolic rule learning	AnyBURL	0.5589	50.78	60.16	0.332*	24.7*	—	1000 s	1000 s*
	AMIE	0.5473	49.58	59.27	0.2170	16.57	23.07	**83 s**	**8 s**
Neural rule learning	NeuralLP	—	—	—	0.3166	24.45	34.31	—	11.9 h
	NCRL	0.380*	27.40*	—	0.300*	20.90*	—	—	—
	RNNLogic	0.5625	49.56	61.80	0.3276	23.49	38.60	4.5 h	0.6 h
Path-based reasoning	PRA / PathRank	0.4678	37.39	55.22	0.0972	6.35	10.72	239 s	229 s
	MINERVA	—	—	—	0.2734	19.77	30.34	—	3.3 h
GNN-based	NBFNet	0.4946	37.96	55.54	**0.5114**	**41.64**	**55.95**	2.3 h	10.4 h
Brain-inspired	NSR	0.5893	52.80	64.32	0.3649	28.45	39.74	0.3 h	0.6 h

*Values reported in prior work or official releases. The AnyBURL FB15k-237 results and its 1000 s rule-learning budget come from the official results page (`https://web.informatik.uni-mannheim.de/AnyBURL/`); NCRL values come from [11]. Published and cross-hardware timings are indicative rather than strictly matched; The reported NSR training time correspond to the accelerated implementation for static knowledge graphs described in Appendix A.2.1.

4.2 Empirical performance

Tables 1 and 2 compare NSR with ten baselines spanning major approaches to knowledge-graph reasoning: embedding models (ConvE [15] and RotatE [48]), symbolic rule mining (AnyBURL [29] and AMIE [20]), path ranking (PRA/PathRank [28]), neural rule learning (NeuralLP [60], NCRL [11], and RNNLogic [39]), reinforcement-learning path search (MINERVA [14]), and graph neural reasoning (NBFNet [62]). We report filtered MRR and Hits@1/3 on Nations, Kinship, YAGO3-10, and FB15k-237. Locally evaluated methods use the same dataset splits and filtered-tail evaluation protocol.

Across the four benchmarks, NSR delivers competitive predictive performance while maintaining strong training efficiency. It ranks near the top on Nations and YAGO3-10 and remains competitive on

FB15k-237, although its performance is weaker on Kinship, indicating dataset-dependent strengths rather than uniformly superior accuracy. At the same time, NSR trains substantially faster than several neural baselines and remains efficient even on the larger graphs. Overall, these results suggest that NSR offers a favorable trade-off between reasoning performance and computational cost, making it a competitive structural reasoner for knowledge-graph tasks where both accuracy and training efficiency matter.

4.3 Traceable reasoning steps

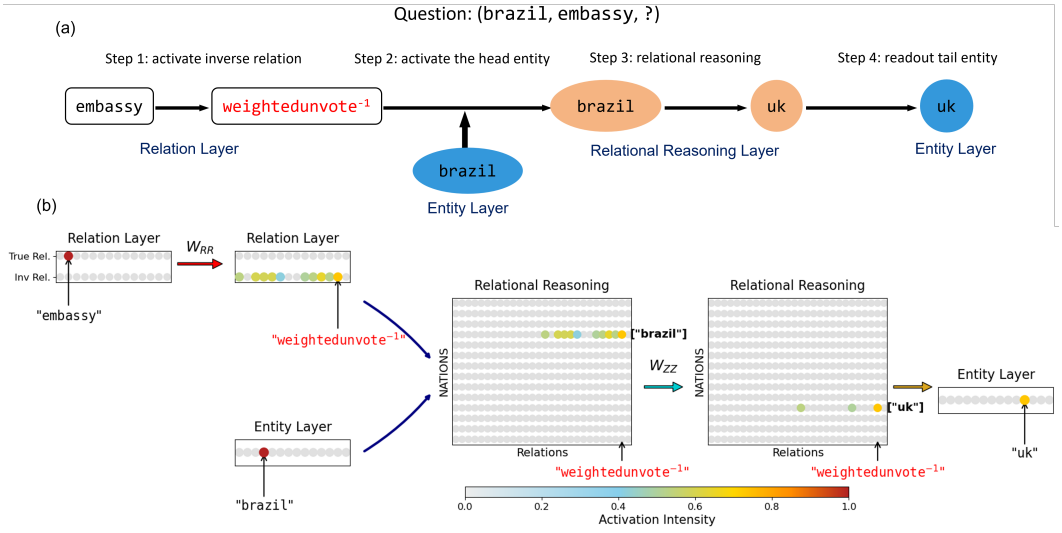


Figure 2: **Traceable reasoning in NSR. (a)** A schematic of the inferred logical path: `embassy` triggers its inverse to map `Brazil` to `UK`. **(b)** Neuronal activations at each stage of the query `(Brazil, embassy, ?)`. Activating the relation neuron `embassy` triggers the activation of candidate equivalent relations (e.g., $\texttt{weightedunvote}^{-1}$ is an associated inverse relation of `embassy`) in the L_R layer. The selected relation and head entity `Brazil` jointly project onto the L_Z layer. State transition in L_Z through recurrent dynamics leads to activation of neurons encoding the tail entity `UK`.

Unlike embedding baselines that rely on post-hoc attention or gradient saliency to explain opaque scalar scores, NSR's interpretability is intrinsic: the activation trajectory *is* the computation. For the query `(Brazil, embassy, ?)` (Figure 2), Step 1 of §3.4 clamps `embassy` onto the L_R layer and iterates the L_R dynamics once through W_{RR}. The Hebbian couplings learned during training (Eq. (4)) activate $\texttt{weightedunvote}^{-1}$ as the dominant equivalent relation, and its activation level serves as the confidence score S_R. Step 2 then activates the head entity `Brazil` in L_E while holding $\texttt{weightedunvote}^{-1}$ active in L_R; the joint input projects onto the L_Z relation map via the $W_{E\leftrightarrow Z}$ and $W_{Z\leftrightarrow R}$, forming the initial state $\boldsymbol{X}_Z^{\text{init}}$. Recurrent dynamics through W_{ZZ} concentrate activation onto the neuron $z_{\text{UK}}^{\text{weightedunvote}^{-1}}$, which is read out in L_E as the tail entity `UK`. Because each active unit represents a discrete, human-readable proposition, errors are auditable by inspecting which relation was activated in L_R or where the L_Z dynamics diverged. The resulting activation sequence provides an inspectable computational trace, inspired by evidence of non-spatial task-state replay in the human hippocampus [45]. Symmetric trace is given in Appendix D.1.

4.4 Learning of latent structures

Latent relational hierarchy. NSR recovers the latent compositional structure of the relational space rather than memorizing pairwise co-occurrences. As shown in Figure 3, the model learns which relation sequences form reliable multi-hop pathways. For example, the chain $\texttt{exportbooks} + \texttt{releconomicaid} \rightarrow \texttt{embassy}$ attains confidence 0.93 because the intermediate transition constitutes a coherent diplomatic pathway, whereas merely repeating `embassy` twice fails to yield a high-confidence rule despite the relation's high frequency. This selectivity shows that composition is

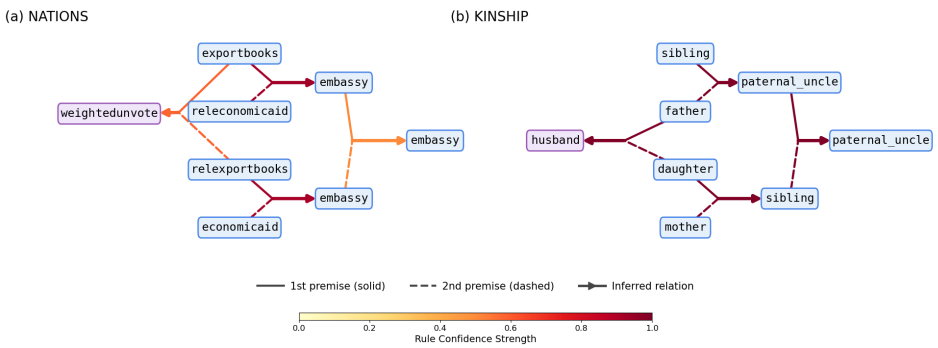


Figure 3: **Latent relational hierarchy.** Compositional rules learned by NSR, satisfying $(h, r_1, x) \wedge (x, r_2, t) = (h, r_3, t)$, on (a) Nations and (b) Kinship1990_EXTENDED. Solid lines denote r_1, dashed lines denote r_2, and color intensity indicates rule confidence.

context-sensitive and depends on whether a relation pair creates a semantically stable bridge. Notably, on Kinship1990_EXTENDED the model can assign confidence 1 to deterministic kinship rules after a single observation by integrating prior logical constraints through hyperparameters, demonstrating that NSR is not limited to statistical induction.

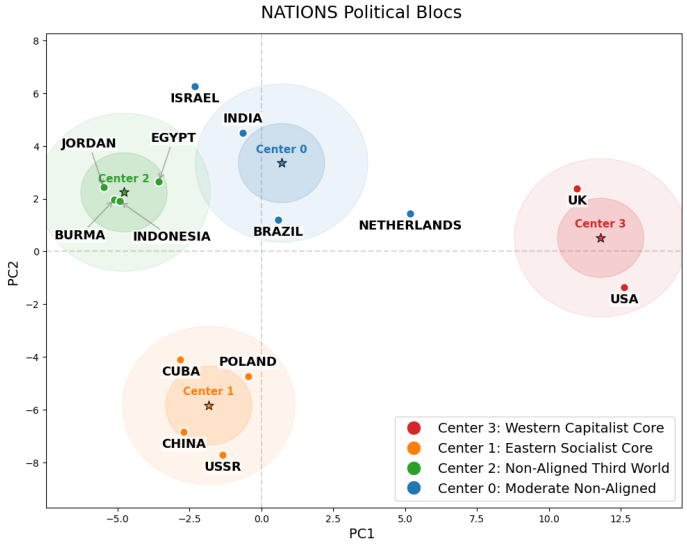


Figure 4: **Learned relational organization in Nations.** PCA projection of nations based on their relational connections in L_Z. Four clusters emerge corresponding to Cold War political blocs.

Relational organization of entities. NSR encodes entities not as isolated symbols but as nodes characterized by their recurrent connections L_Z. As illustrated in Figure 4, projecting entities based on their relational connections (W_{ZZ}) induces structurally meaningful clusters: the Western capitalist core (UK, USA), the Eastern socialist core (USSR, China), and distinct non-aligned blocs (Jordan, Egypt, India, Brazil) each occupy separate regions of the latent space. Taken together, NSR learns a dual latent structure: a compositional hierarchy governing how relations combine, and a relational topology governing how entities are organized.

5 Conclusion

We presented Neural Structural Reasoner (NSR), a brain-inspired architecture that integrates hierarchical representation with relation-driven state transition and Hebbian-style learning to accomplish

structural reasoning over knowledge graphs. By preserving relational structure in layered network connectivity rather than collapsing it into flat embeddings, NSR achieves competitive, dataset-dependent accuracy on standard benchmarks, while its inference dynamics natively expose human-readable reasoning traces. NSR also spontaneously discovers interpretable latent structures such as compositional rules among relations and relational similarity among entities. These properties make NSR a promising step towards building efficient and interpretable models for structural reasoning.

NSR is currently evaluated on discrete triplet-structured knowledge graphs, which provide a controlled setting for structural reasoning but leave open whether the same mechanisms generalize to temporal, hyper-relational, event-based, or noisy real-world knowledge. The model also assumes that entities and relations are already symbolically specified, so future work should couple NSR with neural modules that extract relational structure from text, perception, or episodic experience. While NSR is efficient on current benchmarks, scaling compositional-rule discovery to larger graphs and longer multi-hop chains will require sparse activation, approximate retrieval, or learned proposal mechanisms. Future work should also test robustness to missing or contradictory triples and develop more biologically grounded implementations of sequence detection, gating, and confidence aggregation.

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A Details for Encoding and Learning

A.1 Learning and Encoding of KG Training Data

This subsection details how NSR stores the training graph into its connectivity and extracts pairwise relational structure. Compositional multi-hop rules are deferred to §A.2.

Network initialization. All plastic weights are initialized to zero. The only fixed connections are the bidirectional couplings between each entity neuron e_i and its associated L_Z neurons z_i^k (Eq. (1)).

Encoding and pairwise learning. Each training triple (h_i, r_k, t_j) and its inverse (t_j, r_k^{-1}, h_i) are encoded by setting the corresponding L_Z intra-layer and L_Z–L_R weights to one (Eq. (3)). Concurrently, the network records which relations co-occur on the same (h, t) pair; these co-activations drive the Oja-like updates of W_{RR} (Eq. (4)) that capture equivalence, symmetry, and inverse structure.

Algorithm 1 Learning and encoding of KG training data

Input. Training triples $\mathcal{T}_{\text{train}}$; N entities; M relations.
Parameters. Hebbian rate δ_{RR}, decay μ_{RR}.

initialize all weight matrices to 0
for $i = 1, \ldots, N$ **do**
  for $k = 1, \ldots, 2M$ **do**
    fix $W_{E \leftrightarrow Z}(e_i, z_i^k) = 1$
  end for
end for
for each $(h_i, r_k, t_j) \in \mathcal{T}_{\text{train}}$ **do**
  set $W_{ZZ}(z_i^k, z_j^{k+M}) = W_{ZZ}(z_j^{k+M}, z_i^k) = 1$
  set $W_{Z \leftrightarrow R}(z_i^k, r_k) = W_{Z \leftrightarrow R}(z_j^{k+M}, r_{k+M}) = 1$
  record r_k for pair (h_i, t_j)
end for
for each entity pair (h, t) with ≥ 2 recorded relations **do**
  for each distinct pair (r_i, r_j) in that set **do**
    $\Delta w_{ij}^{RR} \leftarrow \delta_{RR}(r_i r_j) - \mu_{RR}(r_i^2 + r_j^2) w_{ij}^{RR}$
    $w_{ij}^{RR} \leftarrow w_{ij}^{RR} + \Delta w_{ij}^{RR}$
  end for
end for

A.2 Compositional Rules Learning

The Relation Composition layer L_C contains sequence-selective neurons $\{c_{ij}\}$, each tuned to an ordered pair of extended relations $(\tilde{r}_i, \tilde{r}_j)$ with $\tilde{r} \in \{0, \ldots, 2M-1\}$ (forward $\tilde{r}_k \equiv r_k$ for $k < M$, inverse $\tilde{r}_{k+M} \equiv r_k^{-1}$). These neurons detect relational chains through asymmetric-delay coincidence detection.

Detection and encoding of relation composition. When the conditions in (5) are satisfied, assign L_C neurons c_p, c_q and c_k to encode relations r_p, r_q and r_k respectively. In addition, assign neuron $c_{\tilde{p}q}$ to be directly coupled with c_q and indirectly coupled with c_p through an intermediate neuron c_s. The activation threshold of $c_{\tilde{p}q}$ through the sigmoidal nonlinearity renders it responsive only when c_p and c_q are activated in a sequential manner. This enables $c_{\tilde{p}q}$ to serve as a sequence detection neuron in a way that is functionally analogous to direction-selective motion detection models proposed for the visual cortex.

Hebbian learning of compositional rules. When a sequence neuron c_{pq} fires upon detecting the ordered pair (r_p, r_q), and the target relation neuron r_k is concurrently activated (either by a direct triple or by its inverse through the bidirectional encoding of Eq. (3)), the coincident activity strengthens the synapse between c_{pq} and r_k via the same Oja-like rule used in L_R:

$$\Delta w_{k-pq}^{CC} = \delta_{CC}\, c_k\, c_{pq} - \mu_{CC}\,(c_k^2 + c_{pq}^2)\, w_{k-pq}^{CC}. \tag{9}$$

where the decay term effectively controls for the baseline occurrence rate of relations and chains.

The composition-detection and Hebbian mechanisms described above operate locally on individual triple chains. To discover all compositional rules supported by the data, NSR enumerates every two-hop path in $\mathcal{T}$ and applies the closed-loop check of Eq. (5) together with the weight update of Eq. (7). Algorithm 2 implements this global traversal.

Algorithm 2 Compositional-rule learning

Notation. $\mathcal{T}$: training triples (h, r, t) with $r \in \{0, \ldots, M-1\}$; extended relations $\tilde{r} \in \{0, \ldots, 2M-1\}$ encode forward ($\tilde{r}_k \equiv r_k$) and inverse ($\tilde{r}_{k+M} \equiv r_k^{-1}$) links via Eq. (3). c_{pq}: sequence neuron for (r_p, r_q). c_k: neuron in L_C representing relation r_k. δ_{CC}, μ_{CC}: Hebbian rate and decay for L_C synapses.

1: **for** each $(h_i, r_p, x_p) \in \mathcal{T}$ **do**
2: **for** each $(x_p, r_q, x_q) \in \mathcal{T}$ with $r_q \neq r_p^{-1}$ **do**
3: **for** each $(x_q, r_k, h_i) \in \mathcal{T}$ **do**
4: $c_{pq} \leftarrow$ sequence neuron for (r_p, r_q)
5: $c_k \leftarrow$ relation neuron for r_k
6: $\Delta w^{CC}_{k-pq} \leftarrow \delta_{CC}\, c_k\, c_{pq} - \mu_{CC}\,(c_k^2 + c_{pq}^2)\, w^{CC}_{k-pq}$
7: $w^{CC}_{k-pq} \leftarrow w^{CC}_{k-pq} + \Delta w^{CC}_{k-pq}$
8: **end for**
9: **end for**
10: **end for**

A.2.1 Offline learning on static graphs

On a static training graph, the per-experience updates of Algorithm 2 admit an exact closed-form evaluation, so the same connectivity W^{CC} can be computed in a single pass over the encoded weights instead of simulating the network dynamics. Two observations underlie this offline variant.

(i) **Forward episodes reduce to the encoded connectivity.** During composition learning the relation-to-relation couplings W_{RR} are disabled, so an episode initiated at head entity h under relation $\tilde{r}_i$ returns exactly the encoded neighbors of h under $\tilde{r}_i$: the dynamics of Eq. (2) reduce to readout through the fixed connectivity of Eq. (3). Writing $A_{\tilde{r}} \in \{0,1\}^{N \times N}$ for the adjacency matrix of the encoded triples under $\tilde{r}$, the two-hop experiences of a chain $(\tilde{r}_i, \tilde{r}_j)$ are therefore enumerated by the sparse product $S_{ij} = A_{\tilde{r}_i} A_{\tilde{r}_j}$, where $S_{ij}[h, t]$ counts the intermediate entities x with $(h, \tilde{r}_i, x)$ and $(x, \tilde{r}_j, t)$ both encoded.

(ii) **Per-experience updates accumulate in closed form.** For an experience (h, t) of the chain, the return readout has the closed form $y_k(h, t) = A_{r_k}[h, t]$: the neuron c_k is active exactly when (h, r_k, t) is encoded. With a running-mean learning rate $\eta_{ij} = 1/n_{ij}$, the update of lines 6–7 of Algorithm 2 is a running average over experiences, so after all n_{ij} experiences of the chain the weight equals the empirical mean

$$w^{CC}_{k-ij} \;=\; \frac{\langle A_{r_k},\, S_{ij} \rangle_F}{\sum_{h \neq t} S_{ij}[h, t]}, \qquad S_{ij} = A_{\tilde{r}_i} A_{\tilde{r}_j}, \tag{10}$$

where $\langle \cdot, \cdot \rangle_F$ is the Frobenius inner product and the diagonal of S_{ij} is excluded ($t \neq h$). A constant learning rate δ_{CC} yields the standard exponentially weighted estimate of the same quantity when triples arrive sequentially.

Algorithm 3 evaluates Eq. (10) directly. It computes the same weights as the online procedure of Algorithm 2; no discrete rule set is constructed, stored, or re-grounded at query time. The evidence n_{ij} of each chain is returned together with the weights, and the thresholds τ_s and τ_c act only as readout gates during reasoning, in the same way as `T_thresh` gates relation readout in §3.4: a composition neuron c_{ij} participates in reasoning only if $n_{ij} \geq \tau_s$, and a consequent relation r_k is read out from c_{ij} only if $w^{CC}_{k-ij} \geq \tau_c$. The full dynamical implementation and this offline variant attain comparable performance on all benchmarks (§4.2).

Algorithm 3 Offline compositional-rule learning

Notation. $A_{\tilde{r}} \in \{0,1\}^{N \times N}$: adjacency matrix of encoded triples under extended relation $\tilde{r}$ (Eq. (3)). n_{ij}: number of experiences (evidence) of chain $(\tilde{r}_i, \tilde{r}_j)$. **Hyperparameters.** τ_s: minimum evidence for c_{ij} to participate in reasoning (query-time gate). τ_c: minimum weight for reading out a consequent relation (query-time gate).

$W^{CC} \leftarrow 0$
for each ordered pair $(\tilde{r}_i, \tilde{r}_j)$ with $\tilde{r}_j \neq \tilde{r}_i^{-1}$ **do**
  $S_{ij} \leftarrow A_{\tilde{r}_i} A_{\tilde{r}_j}$; set diagonal of S_{ij} to zero *// two-hop experiences*
  $n_{ij} \leftarrow \sum_{h,t} S_{ij}[h,t]$
  if $n_{ij} = 0$ **then continue**
  for $k = 1, \ldots, M$ **do**
    $w^{CC}_{k-ij} \leftarrow \langle A_{r_k}, S_{ij} \rangle_F \,/\, n_{ij}$ *// same weights as Alg. 2*
  end for
end for
return $W^{CC}, \{n_{ij}\}$

B Details for the Reasoning Phase

B.1 Reasoning phase (algorithmic form)

Algorithm 4 restates the five-step procedure of §3.4 in pseudocode. Layer updates follow the dynamics defined in Eq. (2); the final aggregation follows Eq. (8).

B.2 Hyperparameter settings

Table 3 lists the hyperparameters of the simplified variant used in §3. They are grouped by the processing stage to which they belong.

Reasoning threshold. `T_thresh` is the activation threshold of Step 1 (§3.4): a relation neuron r_e enters the equivalent-relation set $\mathcal{R}_{\text{eq}}$ only if its activation satisfies $x_e \geq$ `T_thresh`. Given that the Countries dataset contains only two relations, reasoning does not depend on relation similarity; hence, `T_thresh` is set to infinity.

Learning rate. The simplified implementation collapses the Hebbian rates δ_{RR} (Eq. (4)) and δ_{CC} (Eq. (7)) to a single scalar δ_θ, because the simplified variant performs one-pass counting instead of iterative Oja updates. The value reported in Table 3 therefore serves as the effective learning rate for both W_{RR} and W_{CC}.

Compositional-rule thresholds. τ_s and τ_c are the query-time readout gates of Algorithm 3: during reasoning, a composition neuron c_{ij} participates only if its evidence n_{ij} is at least τ_s, and a consequent relation is read out from c_{ij} only if its weight w^{CC}_{k-ij} is at least τ_c. All values were obtained with `Optuna` [1]; the search ranges were `T_thresh` $\in [0.05, 1.0]$, $\delta_\theta \in [0.1, 1.0]$, $\tau_s \in [1, 50]$, and $\tau_c \in [0.05, 0.5]$. NSR is not highly sensitive to these settings.

Table 3: Hyperparameters for the main KG benchmarks

Dataset	`T_thresh`	τ_s	τ_c	δ_θ
Nations	0.22	23	0.20	0.52
Kinship	0.90	27	0.17	0.20
Countries S3	inf	1	1.00	1.00
Kinship1990_EXTENDED	0.46	23	0.13	0.34

Algorithm 4 Reasoning phase for query $(h_i, r_k, ?)$ with max/sum aggregation.

Require: Query head h_i, query relation r_k; trained network weights and layer dynamics in Eq. (2); activation threshold `T_thresh`; gate function $G = \mathrm{TopK}$; aggregation mode $a \in \{\max, \mathrm{sum}\}$, selected on the validation set.

Ensure: Candidate-tail ranking scores $\mathrm{Score}_a(t)$.

Initialize path collections $\mathcal{P}_R \leftarrow \emptyset$ and $\mathcal{P}_C \leftarrow \emptyset$.
{**Step 1:** Retrieve equivalent relations from L_R}
Initialize all layer activations $X_E, X_Z, X_R, X_C \leftarrow \mathbf{0}$.
Clamp query relation: $X_R[r_k] \leftarrow 1$.
Update L_R once using Eq. (2).
$\mathcal{R}_{\mathrm{eq}} \leftarrow \{r_e \mid X_R[r_e] \geq \texttt{T_thresh}\}$.
Store $S_R(r_e) \leftarrow X_R[r_e]$ for each $r_e \in \mathcal{R}_{\mathrm{eq}}$.
{**Step 2:** Relational reasoning via L_Z}
Reset $X_E, X_Z \leftarrow \mathbf{0}$; retain only $\mathcal{R}_{\mathrm{eq}}$ in L_R.
Activate the head entity: $X_E[e_i] \leftarrow 1$.
Iterate L_Z and read out L_E using Eq. (2).
Collect the grounded one-hop paths $p = (h_i, r_e, t)$ realized by these transitions into $\mathcal{P}_R$.
for each $p = (h_i, r_e, t) \in \mathcal{P}_R$ **do**
    Assign path-support score $s(p) \leftarrow S_R(r_e)$.
end for
{**Step 3:** Retrieve relation compositions from L_C}
Re-initialize $X_E, X_Z, X_R, X_C \leftarrow \mathbf{0}$.
Clamp query relation: $X_R[r_k] \leftarrow 1$.
Propagate activity to L_C and update its dynamics using Eq. (2).
$\mathcal{C}_{\mathrm{eq}} \leftarrow G(X_C) = \mathrm{TopK}(X_C)$.
Store $S_C(c_e) \leftarrow X_C[c_e]$ for each $c_e \in \mathcal{C}_{\mathrm{eq}}$.
{**Step 4:** Multi-hop reasoning with each composition}
for each $c_e \in \mathcal{C}_{\mathrm{eq}}$ **do**
    Reset $X_E, X_Z, X_R, X_C \leftarrow \mathbf{0}$.
    Activate c_e in L_C and initialize L_E at e_i.
    Decode c_e into the ordered relation chain $(r_{p_1}, r_{p_2}, \ldots, r_{p_\ell})$.
    for $j = 1, \ldots, \ell$ **do**
        Activate r_{p_j} in L_R.
        Iterate L_Z and read out L_E using Eq. (2), retaining distinct grounded transition traces.
        Use the reached entities as the starting states for the next hop, if any.
    end for
    Collect the completed grounded paths $p = (h_i, r_{p_1}, x_1, \ldots, r_{p_\ell}, t)$ into $\mathcal{P}_{c_e}$.
    for each $p \in \mathcal{P}_{c_e}$ **do**
        Assign path-support score $s(p) \leftarrow S_C(c_e)$.
    end for
    $\mathcal{P}_C \leftarrow \mathcal{P}_C \cup \mathcal{P}_{c_e}$.
end for
{**Step 5:** Max/sum aggregation of path-support scores}
$\mathcal{P} \leftarrow \mathcal{P}_R \cup \mathcal{P}_C$.
$\tilde{\mathcal{T}} \leftarrow \{t \mid \exists p \in \mathcal{P}$ with terminal entity $t\}$.
for each $t \in \tilde{\mathcal{T}}$ **do**
    $\mathcal{P}(t) \leftarrow \{p \in \mathcal{P} \mid p$ has terminal entity $t\}$.
    if $a = \max$ **then**
        $\mathrm{Score}_a(t) \leftarrow \max_{p \in \mathcal{P}(t)} s(p)$.
    else
        $\mathrm{Score}_a(t) \leftarrow \sum_{p \in \mathcal{P}(t)} s(p)$.
    end if
end for
return Score_a.

C Datasets and Detailed Experiments

C.1 Datasets

We evaluate NSR on knowledge-graph datasets of varying scale and relational complexity. Nations, Kinship, and Countries S3 are standard benchmarks in the KG-embedding literature. Kinship1990_EXTENDED is an extended derivative of the classical Kinship domain that we introduce to stress-test compositional relational reasoning; its construction is detailed in § C.2. WN18RR, FB15k-237, and YAGO3-10 are standard large-scale benchmarks. Table 4 summarizes the number of entities, relation types, and triples (edges) in each split.

Table 4: Statistics of the datasets used in our experiments. All counts in the last five columns refer to the number of edges (triples).

Dataset	Entities	Relations	Total	Train	Val	Test
Nations	14	55	1,992	1,592	199	201
Kinship	104	26	10,686	8,544	1,068	1,074
Kinship1990_EXTENDED	480	14	2,240	1,568	224	448
Countries_S3	271	2	1,033	985	24	24
WN18RR	40,943	11	93,003	86,835	3,034	3,134
FB15k-237	14,541	237	310,116	272,115	17,535	20,466
YAGO3-10	123,182	37	1,089,040	1,079,040	5,000	5,000

C.2 How to Construct Kinship1990_EXTENDED

Original data and its limitations. The original Kinship dataset [33] contains 112 triples over 24 persons from two isomorphic families, with 12 binary relations: `wife`, `husband`, `mother`, `father`, `daughter`, `son`, `sister`, `brother`, `aunt`, `uncle`, `niece`, and `nephew`. Three features limit its utility for testing compositional reasoning: (i) *No symmetric relations*—every relation is directional, so link-prediction can be solved by learning inverses alone; (ii) *Flat relation labels*—genealogical concepts such as `aunt` or `uncle` are provided as atomic symbols rather than explicit compositions of primitives (e.g., $\texttt{mother} \circ \texttt{sister}$), so models are not required to discover relational grammar; (iii) *Small scale*—112 triples provide insufficient coverage to stress-test generalization.

Reconstructing the base graph. Because the two families have identical structure, the complete genealogical tree is fully determined by the original triples. We first reconstruct this tree, obtaining all parent–child and spousal links.

Injecting symmetric relations. We add `sibling` as an explicitly symmetric relation. For every pair of distinct children sharing at least one parent in the reconstructed tree, we insert $\texttt{sibling}(X, Y)$ and $\texttt{sibling}(Y, X)$. Unlike the original gender-specific `brother` and `sister`, `sibling` forces the model to respect an equivalence constraint that cannot be reduced to a directional inverse.

Extracting compositional chain relations. We define five new relation types by enumerating multi-hop genealogical paths in the reconstructed tree and abstracting them into explicit labels:

- `grandmother_chain`: the mother of one's father or mother;
- `grandfather_chain`: the father of one's father or mother;
- `maternal_aunt_chain`: the sister of one's mother;
- `paternal_uncle_chain`: the brother of one's father;
- `cousin_chain`: the child of one's parent's sibling.

For example, $\texttt{cousin_chain}(h, t)$ is generated whenever there exist intermediates x, y such that x is a parent of h, y is a sibling of x, and t is a child of y. These chain relations are not redundant with the original flat labels; they make compositional substructure explicit and therefore require the model to reuse primitive relations rather than memorize atomic mappings.

Statistics and splits. The extended graph contains 480 entities, 14 relation types, and 2,240 triples. We follow the protocol: 70% train, 10% validation, 20% test. Because the underlying family tree is small and structurally deterministic, all base-relation edges necessary for genealogical consistency appear in the training split; held-out test triples are drawn predominantly from the extended relation set so that accurate prediction benefits from compositional reuse of the trained primitives.

C.3 Detailed Experiments

Embedding-Based baseline implementations. All neural baselines are trained with PyKEEN v1.10 [2] under identical data splits. Table 5 lists the architecture and optimization settings shared across embedding models. The embedding dimension is set to 100 for small-scale datasets (e.g., Nations and Kinship) and to 200 for large-scale datasets (e.g., WN18RR, FB15k-237 and YAGO3-10). DistMult, ComplEx, RESCAL, and ConvE use the LCWA training loop; TransE and RotatE use sLCWA. ConvE requires inverse triples (create_inverse=True) per PyKEEN's implementation and uses 32 output channels with dropout rates $\{0.2, 0.2, 0.3\}$. The optimizer is Adam with learning rate 10^{-3} for all models except RotatE (5×10^{-4}). Training runs for at most 150 epochs with batch size 32.

Table 5: Neural baseline hyperparameters and training configuration.

Model	Dim	Loop	Inverse	LR	Epochs	Special
TransE	100	sLCWA	No	1e−3	150	scoring_fct_norm=1
DistMult	100	LCWA	No	1e−3	150	—
ComplEx	100	LCWA	No	1e−3	150	—
RotatE	100	sLCWA	No	5e−4	150	—
ConvE	100,200	LCWA	Yes	1e−3	150	out_ch=32, dropouts $\{0.2, 0.2, 0.3\}$
RESCAL	100	LCWA	No	1e−3	150	—

Rule Learning baseline implementations. RNNLogic operates *without pretrained knowledge-graph embeddings*. It first mines relational paths up to length 3 using the C++ miner (16 threads), then learns rule weights via a lightweight predictor trained for 10 iterations (Adam, learning rate 5×10^{-3}, weight decay 0, hidden dimension 32, batch size 16). Label smoothing (0.1) is applied during training, and expectation-based ranking (`expectation=True`) is used at evaluation. Unlike neural baselines, no entity or relation embeddings are supplied or learned.

Reproducibility. We used five random seeds: [42, 43, 44, 45, 46], and the performance of these baseline models and NSR is reported in § 4.2. PyKEEN handles its own internal seeding for negative sampling and parameter initialization. NSR and all baselines are evaluated on identical train/validation/test splits for each dataset.

Table 6: Symbolic baseline hyperparameters and training configuration.

Model	Rule Len	Hidden Dim	LR	Iters	Special
RNNLogic (Miner)	3	—	—	—	16 threads
RNNLogic (Predictor)	—	32	5e−3	10	batch=16, smoothing=0.1, expectation=True, **without embedding**

C.4 Additional Benchmark Results

We report three complementary evaluations deferred from the main text: Countries S3, which probes multi-hop chain generalization; Kinship1990_EXTENDED, our controlled compositional stress test (§C.2); and WN18RR, a standard large-scale benchmark. We then compare NSR against eight symbolic, neuro-symbolic, and neural reasoning baselines across all five benchmarks, and report training and inference costs.

Countries S3. Test queries ask for a country's continent, but training triples only link neighboring countries and upward `locatedIn` edges; no country–continent fact is observed directly, so answering

requires composing `neighborOf` and `locatedIn` chains. Table 7 shows that NSR solves the task perfectly, while embedding baselines collapse (best MRR 0.199) because static vector similarity cannot recover unobserved multi-step paths, and RNNLogic reaches only 0.341 because externally mined templates miss the compositional regularity.

Table 7: Performance comparison on Countries S3. Best results in each column are highlighted in bold. Hits@k are reported as percentage values in $[0, 100]$; standard deviations are in units of 10^{-2} (MRR) and percentage points (Hits@k).

Model	MRR ↑	Hits@1 ↑	Hits@3 ↑	**Train Time (s)**
ConvE	0.1987±2.47	11.25±2.50	19.17±4.04	20.3±2.7
DistMult	0.1761±1.42	7.50±1.02	20.42±3.06	17.4±1.0
RotatE	0.1184±2.42	4.17±1.86	10.00±4.45	19.4±2.9
RESCAL	0.1906±1.00	10.42±2.64	18.33±2.04	14.7±2.1
ComplEx	0.0290±2.68	0.83±1.67	2.08±4.17	18.4±2.2
TransE	0.1182±0.61	0.00±0.00	14.17±2.76	18.9±1.7
RNNLogic	0.3409±3.29	0.00±0.00	62.50±12.36	5.1±0.0
NSR	**1.0000**±0.00	**100.00**±0.00	**100.00**±0.00	**0.0012**±0.0004

Kinship1990_EXTENDED. On the composition-focused extension, test triples are drawn predominantly from the five composition-defined chain relations (§C.2). NSR attains the best MRR (0.953) and Hits@1 (94.3), surpassing all baselines including DistMult, which is strong on MRR (0.941) but relies on memorized symmetric structure; the inversion relative to classical Kinship, where ConvE leads, is discussed in §4.2.

Table 8: Performance comparison on Kinship1990_EXTENDED. Conventions as in Table 7.

Model	MRR ↑	Hits@1 ↑	Hits@3 ↑	**Train Time (s)**
ConvE	0.8265±0.91	74.93±1.47	87.83±0.72	60.5±7.5
DistMult	0.9409±0.15	90.62±0.30	**97.75**±0.25	31.9±4.0
RotatE	0.5988±4.47	51.94±5.27	64.40±4.20	26.0±1.1
RESCAL	0.0144±0.28	0.22±0.12	0.80±0.31	30.3±5.3
ComplEx	0.0220±0.34	0.42±0.23	1.41±0.50	31.4±1.6
TransE	0.2249±0.58	2.19±0.69	33.42±1.58	25.3±2.4
RNNLogic	0.8107±1.82	80.45±1.92	81.38±1.78	9.3±1.4
NSR	**0.9533**±0.30	**94.33**±0.48	96.47±0.09	**0.0451**±0.0002

WN18RR. Table 9 reports full filtered metrics on WN18RR, together with the symbolic and neuro-symbolic baselines (MRR; see Table 10 for the remaining datasets). NSR attains the best Hits@1 (0.463) among all evaluated methods and is competitive with the strongest embedding models on MRR, while the GNN reasoner NBFNet remains stronger; we view NSR as competitive rather than uniformly superior on this benchmark.

Comparison with symbolic and neuro-symbolic reasoners. Table 10 compares NSR against eight reasoning baselines spanning symbolic rule mining (AnyBURL, AMIE), path ranking (PRA/PathRank), differentiable rule learning (NTP, NeuralLP, NCRL), RL-based path search (MINERVA), and GNN reasoning (NBFNet) on all five benchmarks. NSR outperforms the evaluated differentiable rule learners on every dataset they complete, performs comparably to strong symbolic rule miners overall, and achieves the best result on YAGO3-10, while NBFNet remains stronger on WN18RR and FB15k-237 and AMIE is strongest on Nations. Because the three large benchmarks are heterogeneous and not designed around NSR's compositional structure, these results indicate that NSR's performance is not confined to small, composition-aligned benchmarks.

Training and inference cost. Table 11 reports end-to-end training time and full-test-set inference time. NSR trains in seconds to minutes across all benchmarks—orders of magnitude below iterative neural training on the large datasets—while inference cost remains modest; measurements on baselines span different hardware and stopping criteria, so we report them as indicative rather than as strict multiplicative speedups.

Table 9: Filtered link prediction on WN18RR. Symbolic and neuro-symbolic baselines report MRR only; "—" indicates unreported or not scalable. Best result in each column in bold.

Method	MRR	Hits@1	Hits@3	Hits@10
TransE	.226	—	—	.501
DistMult	.430	.390	.440	.490
ConvE	.430	.400	.440	.520
ComplEx	.440	.410	.460	.510
RotatE	.476	.428	.492	.571
BoxE	.451	.400	.472	.541
ModE	.472	.427	.486	.564
HAKE	.497	.452	**.516**	.582
SectorE	.475	.421	.478	**.586**
AnyBURL	.5658	—	—	—
AMIE	.4157	—	—	—
PRA / PathRank	.0556	—	—	—
NTP	—	—	—	—
NeuralLP	.4677	—	—	—
NCRL	.4070	—	—	—
MINERVA	.4890	—	—	—
NBFNet	**.5976**	—	—	—
NSR	.472	**.463**	.483	.484

Table 10: Link prediction results (MRR) on all KG completion benchmarks. Best results are in **bold**; asterisks (*) denote values taken from prior work or official repositories, and "—" indicates results not reported or the method does not scale to the dataset.

Method	**Nations**	**Kinship**	**WN18RR**	**YAGO3-10**	**FB15k-237**
AnyBURL	0.7994	0.6768	0.5658	0.5589	0.332*
AMIE	**0.8559**	0.6767	0.4157	0.5473	0.2170
PRA / PathRank	0.5933	0.6296	0.0556	0.4678	0.0972
NTP	0.6223	0.612*	—	—	—
NeuralLP	0.6841	0.6072	0.4677	—	0.3166
NCRL	0.4571	0.6050	0.4070	0.380*	0.300*
MINERVA	0.5865	0.6253	0.4890	—	0.2734
NBFNet	0.7479	**0.7445**	**0.5976**	0.4946	**0.5114**
NSR	0.8142	0.6515	0.4720	**0.5893**	0.3649

Table 11: Training time / full-test-set inference time (seconds unless noted). **quoted from the official AnyBURL-23-1 release; other local runs use the same machine. NBFNet entries use official training profiles, so cross-hardware comparisons are indicative. OOM: out of memory.

Method	**Nations**	**Kinship**	**WN18RR**	**YAGO3-10**	**FB15k-237**
AnyBURL	64 s / 16 s	63 s / 36 s	603 s / 11 s	1000 s / 424 s	1000 s** / —
AMIE	2246 s / <0.1 s	2 s / 0.3 s	2 s / <0.1 s	83 s / 0.6 s	8 s / 1.2 s
PRA / PathRank	34 s / 0.5 s	86 s / 0.3 s	22 s / 156 s	239 s / 354 s	229 s / 841 s
NTP	1.9 h / 8 s	—	—	—	—
NeuralLP	40 s / 8 s	26 s / 4 s	1.1 h / 129 s	— (OOM)	11.9 h / 533 s
NCRL	134 s / 43 s	99 s / 7 s	244 s / 24 s	—	—
MINERVA	3160 s / 8 s	1.2 h / 13 s	3416 s / 44 s	—	3.3 h / 333 s
NBFNet	29 s / 0.1 s	77 s / 0.3 s	5.6 h / 12 s	2.3 h / 422 s	10.4 h / 46 s
NSR	3 s / 0.4 s	11 s / 1.2 s	3.3 s / 32 s	1142 s / 107 s	2019 s / 55 s

C.5 Ablation Study

We ablate the four learnable or structural components of NSR to isolate their individual contributions: inverse-relation encoding (Eq. (3)), relation-equivalence retrieval (Step 1 of §3.4), compositional inference (Steps 3–4 of §3.4), and Hebbian learning (Eqs. (4) and (7)). All ablations are conducted with the full dynamical model on Nations, Kinship, and Kinship1990_EXTENDED, under the filtered link-prediction protocol of §4.1. Each variant is evaluated on five paired seeds; we report MRR (mean $\pm$ population standard deviation across seeds) together with ΔMRR relative to the paired full model. Ties between equally scored entities are broken by seed-dependent uniform noise of magnitude 10^{-6}.

Table 12: Component-wise ablations of NSR (MRR, mean $\pm$ std over five seeds). Values in parentheses denote ΔMRR relative to the paired full model. Best results in each column are highlighted in bold.

Variant	Nations	Kinship	Kinship1990_EXTENDED
Full model	**0.81**$\pm$0.03	**0.65**$\pm$0.01	**0.95**$\pm$0.00
w/o inverse encoding	0.66$\pm$0.03 (-0.15)	0.05$\pm$0.00 (-0.60)	0.10$\pm$0.00 (-0.85)
w/o relation-equivalence retrieval	0.61$\pm$0.01 (-0.20)	0.42$\pm$0.00 (-0.24)	0.82$\pm$0.00 (-0.12)
w/o compositional inference	0.79$\pm$0.03 (-0.03)	0.48$\pm$0.01 (-0.17)	0.46$\pm$0.00 (-0.48)
w/o Hebbian learning	0.36$\pm$0.01 (-0.45)	0.05$\pm$0.00 (-0.60)	0.02$\pm$0.00 (-0.93)

The variants are defined as follows. *W/o inverse encoding*: the model is retrained from scratch with the inverse half of the encoding in Eq. (3) disabled—inverse triples (t, r^{-1}, h) are never encoded and the inverse relation neurons $\{r_{i+M}\}$ of L_R are unused; both learning phases are otherwise unchanged. *W/o relation-equivalence retrieval*: training is identical to the full model and the same trained network is reused; only the reasoning phase is modified, with the W_{RR} propagation of Step 1 in §3.4 disabled at query time, so a query is answered using only the query relation itself and its associated compositions. *W/o compositional inference*: again the same trained network as the full model is reused, and the L_C pathway (Steps 3–4 of §3.4) is disabled at query time, so candidate tails come exclusively from equivalent-relation paths $\tilde{\mathcal{T}}^{(1)}$. *W/o Hebbian learning*: the encoding phase (Eq. (3)) proceeds as in the full model, but both Hebbian learning phases are skipped, so W_{RR} and W^{CC} remain at their zero initialization; the reasoning pipeline is run unchanged.

Findings. The degradation pattern is functionally specific rather than uniform, showing that each component serves a distinct role. Inverse encoding is decisive on the kinship benchmarks (-0.60 and -0.85 MRR): directed kinship relations such as `son_of` or `grandmother_chain` can only be traversed through their inverses, whereas Nations (-0.15) usually offers an equivalent forward relation as a detour. Relation-equivalence retrieval contributes consistently across all datasets (-0.12 to -0.24), confirming that the learned W_{RR} couplings provide a reliable first source of candidate relations at query time. Compositional inference matters most on Kinship1990_EXTENDED (-0.48), whose test queries require unseen multi-hop compositions by construction, and least on Nations (-0.03), where most test queries are solvable by single-hop equivalence—mirroring the intended difference between the two benchmarks. Finally, disabling Hebbian learning causes the largest overall degradation (-0.45, -0.60, -0.93), showing that the learned associative connectivity is essential to performance rather than a merely decorative biological motif. The small residual MRR of this variant is expected: with W_{RR} and W^{CC} at zero, no equivalent relation or composition can be retrieved, entity scores degenerate, and ranks are effectively determined by tie-breaking noise, whose expected reciprocal rank is nonzero on small entity sets.

We stress the scope of this evidence: these ablations validate the *computational* role of each implemented component. They are not evidence for or against the biological realism of the corresponding mechanisms, which we present as computational inspiration.

Aggregation rule. We also ablate the max aggregation of Eq. (8) against sum aggregation. The two rules exhibit a genuine trade-off: max aggregation can be dominated by a single strong but spurious path, whereas sum aggregation pools convergent evidence but may overcount correlated paths. For example, on the Kinship query (`person80`, `term16`, ?), the target `person25` has no direct evidence; under max aggregation a competitor (`person32`, score 0.662) slightly outscores the target (0.641), while under sum aggregation the target accumulates support from 132 grounded paths (total 46.23, versus 1.55 from 3 paths for the competitor) and becomes the unique top-ranked answer. We

therefore select the aggregation mode on validation data and treat NSR's outputs as ranking scores rather than calibrated probabilities.

D Additional Analyses and Reasoning Traces

D.1 Symmetry Reasoning Trace

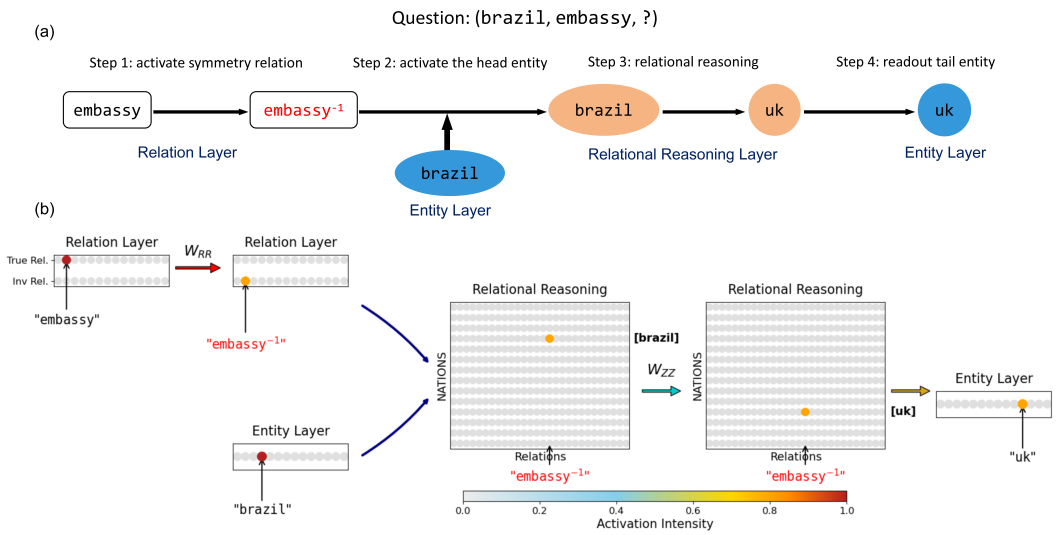


Figure 5: **Symmetry-reasoning trace (symmetry branch only).** In Path A, symmetry and inverse evaluations are performed in parallel. The complete trace therefore contains two branches: a symmetry branch that retains the original relation neuron, and an inverse branch that activates the inverse-relation neuron. For visual clarity, only the symmetry branch is shown here; the inverse branch follows the same gated-diffusion and binding structure with relation inversion.

Figure 5 illustrates the activation trajectory for symmetry reasoning on the query `(Brazil, embassy, ?)`. During the associative phase, the query relation `embassy` diffuses through W_{RR}, and the complementary gate promotes both symmetric and inverse counterparts in parallel. The full inference thus comprises two structurally isomorphic branches: one retaining the original relation `embassy` (symmetry) and one activating $\texttt{embassy}^{-1}$ (inverse). Because the two branches differ only in whether the relation neuron is inverted, we display only the symmetry branch for clarity; the inverse branch can be obtained by following the same pipeline with the W_{RR} dynamics.

D.2 Latent Relational Structure of Countries

The Countries dataset is built from only two primitive relations, `locatedIn` and `neighborOf`, distributed over three entity strata: countries, regions, and continents. Crucially, `neighborOf` edges are observed *only* between country-level entities; the dataset contains no direct adjacency between regions or between continents, even though such adjacencies are clearly entailed by the underlying geography. Recovering these missing higher-order neighborhoods therefore provides a natural probe of NSR's compositional generalization, requiring the model to abstract the same adjacency relation across the entity hierarchy rather than within a fixed layer.

Two chained rules suffice to lift adjacency upward across strata:

$$\texttt{nb}(c_a, c_b) \wedge \texttt{loc}(c_a, R_a) \wedge \texttt{loc}(c_b, R_b) \Rightarrow \texttt{nb}(R_a, R_b), \tag{11}$$

$$\texttt{nb}(R_a, R_b) \wedge \texttt{loc}(R_a, K_a) \wedge \texttt{loc}(R_b, K_b) \Rightarrow \texttt{nb}(K_a, K_b), \tag{12}$$

where $\texttt{nb} \equiv \texttt{neighborOf}$ and $\texttt{loc} \equiv \texttt{locatedIn}$; $c_\bullet$, $R_\bullet$, and $K_\bullet$ index countries, regions, and continents respectively.

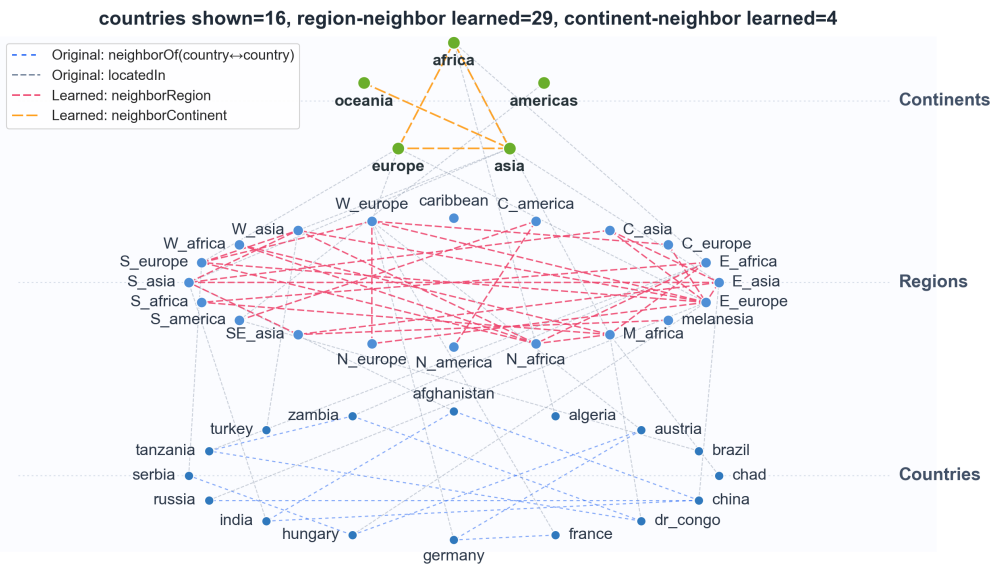


Figure 6: **Discovery of latent `neighborOf` on Countries.** Blue and gray dashed edges denote the original `neighborOf` and `locatedIn` facts; red and orange dashed edges denote the region- and continent-level adjacencies induced by NSR.Region prefixes abbreviate cardinal directions: N (northern), S (southern), E (eastern), W (western), C (central), M (middle), SE (southeastern). Only the 16 most active countries are shown; all regions and continents are displayed in full.

Figure 6 renders the dataset as a three-tier layered graph. Without ever observing a single direct edge of either type during training, NSR recovers 29 region-neighbor and 4 continent-neighbor edges, and the induced adjacencies are geographically coherent: the continent layer assembles the expected Africa–Europe–Asia triangle together with an Asia–Oceania link, while the region layer reproduces dense intra-Europe, intra-Africa, and intra-Asia neighborhoods (e.g., `western_europe` adjoining `northern_europe` and `southern_europe`).